The normalised gaps of the scale-invariant point process follow a GEM distribution

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Editorial note. The counted statement is the author's labelled prop environment at source0.tex lines 706-709; its proof environment opens at line 711, the line after the blank separator, so the pairing is by adjacency and the retained proof range 711-713 is the whole environment.

Editorial note. Two statements are retained as context because the counted proof depends on them: the scale-invariant spacing lemma (source lines 695-698), which describes the process being transformed, and the mapping theorem prel:map (source lines 1805-1808), which the proof invokes at source line 712.

Editorial note. This article carries no label-number dump and the retained counter values are left empty for the whole source, so the printed numbering of the card is the wrapper's default; the identity of the counted item is its author label prop:gemfrompi.

Editorial scope. One original result of the pinned publisher article, counted once: the normalised gaps of the scale-invariant point process follow a GEM distribution (author label prop:gemfrompi).

Allowed standard prerequisites (editorial).

- Scale-invariant spacing process. The points S_i indexed by the integers form a Poisson process on the positive half-line with intensity ds/s.. It is the process transformed in the proof.
- Mapping theorem for Poisson processes. A measurable image of a Poisson process is again a Poisson process with the pushed-forward mean measure, provided that measure has no atoms.. It transports the process to the real line.
- GEM distribution. The GEM distribution is the law of the decreasing rearrangement of the atoms obtained by cutting a unit interval at independent geometric proportions.. It is the law asserted for the normalised gaps.

Lemma 0.1 (Scale-invariant spacing lemma)**.** *The point process* $\{S_i : i \in \mathbb{Z}\}$ *is a Poisson process on the positive real line with intensity measure* $\mathrm{d}s/s$.

Proposition 0.2 (Mapping theorem, [11, §2.3]). *If* Π *is a Poisson process with mean measure* μ *on* S*, and* $f : S \to T$ *is a measurable function such that* $\mu^*(B) \coloneqq \mu(f^{-1}(B))$ *has no atoms, then* $f(\Pi)$ *is a Poisson process on* T *with measure* μ^**.*

Proposition 0.3 (Arratia, [2]). *The process* $\left(1 - \frac{S_1}{\log x}, \frac{Y_1}{\log x}, \frac{Y_2}{\log x}, \dots\right)$ *follows a GEM distribution.*

Proof. Applying the map $T_x(s) \coloneqq \log_2 x - \log s$ to the points of $\{S_i : i \in \mathbb{Z}\}$ yields a homogeneous Poisson process on the real line with constant rate 1, by the mapping theorem (Proposition 0.2) and Lemma 0.1. Furthermore, we have $T_x(S_0) < 0 \leq T_x(S_1)$ and $(T_x(S_i))_{i\in\mathbb{Z}}$ increasing. Therefore, $T_x(S_1)$ and $T_x(S_{i+1}) - T_x(S_i)$ for all $i \geq 1$ are independent exponential random variables of parameter 1. If X is a standard exponential random variable, then $1 - e^{-X}$ is a uniform random variable in $[0, 1]$. We conclude that $1 - {}^{S_1}/_{\log x}$, ${}^{Y_1}/_{S_1}$, ${}^{Y_2}/_{S_2}, \dots$ are independent uniform random variables in $[0, 1]$. The proposition follows by the characterization of the GEM distribution described in the introduction. □

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